

Entropic Redshift and the RSVP Stack: A Unified Field Ontology for Cosmos, Mind, and Society

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The universe is not made of things, but of processes. A. N. Whitehead

The Fragmentation of Reality

Cosmology

- Metric expansion
- Dark energy
- Redshift = z_{Λ}

Cognition

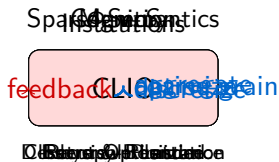
- Symbolic computation
- Free energy principle
- Sparse coding

Society

- Institutions as contracts
- Memetic evolution
- Boundary rituals

No shared dynamical language.

The RSVP Stack: Unified Field Ontology



All layers = functors on same field dynamics.

RSVP Field Equations

From variational principle:

$$\mathcal{A} + \mathcal{D} = \int \left(\frac{1}{2} \|\mathbf{v}\|^2 + \frac{1}{2} \Phi^2 - S \right) + \text{dissipation}$$

$$\partial_t \Phi = -\nabla \cdot (\Phi \mathbf{v}) + \kappa_\Phi \Delta \Phi$$

$$\partial_t \mathbf{v} = -(\mathbf{v} \cdot \nabla) \mathbf{v} - \nabla \Phi + \kappa_v \Delta \mathbf{v}$$

$$\partial_t S = \sigma + \kappa_S \Delta S - \lambda S$$

Interpretation

- Φ : semantic density
- $\mathbf{v}$: agency flow
- S : entropic cost

Entropic Redshift: No Expansion Needed

$$z_{\text{RSVP}} = \exp \left(\alpha_S \int \partial_t S dt \right) - 1$$

- Redshift = cumulative entropy processing
- Monotonic in time
- Scales as $\sim \int \sigma dt / R(t)^2$

redshift_plot.png

phi_3d.png

Void Formation

vorticity_3d.png

Vortex Persistence

Performance

- 16M points/field $\times$ 5 fields

Three Falsifiable Predictions

Domain	Prediction
Cosmology	z_{RSVP} fits DESI/Pantheon+ better than ΛCDM
Neuroscience	$\Gamma_{\text{fMRI}} \propto$ working memory persistence ($r > 0.6$)
Sociology	Spherepop merge rate matches historical data (1870–2020)

Reject if $\Delta\chi^2 > 25$ (cosmology) or $p < 0.001$ (others).

Universal Collapse Patterns

Layer	Failure
RSVP	$\nabla\Phi \rightarrow 0, \mathbf{v} \rightarrow 0, S \rightarrow B_S$
EBSSC	Over-dense semantics
Spherepop	Boundary dissolution
TARTAN	Tile decoherence
CLIO	Gradient vanishing

Same math $\rightarrow$ same interventions: add noise, harden boundaries.

Next Steps

- ① **Nonlocal RSVP**: Fractional Laplacian for long-range semantics
- ② **Quantum RSVP**: Fields on Fock space
- ③ **Real-time CLIO**: Neural fields for live inference
- ④ **Data Fusion**: Joint fit to DESI + fMRI + merger data

One field. One dynamics. All scales.

The RSVP stack is:

- **Mathematically rigorous** (weak solutions, functors)
- **Computationally scalable** (TPU 256^3)
- **Empirically falsifiable** (3 domains)

Lets unify science.

Thank You

Questions? Code, data, simulations, papers